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CEREBELLAR MODEL CONTROL OF AN
INDUSTRIAL ROBOT MANIPULATOR

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Abstract

In this paper, a general learning algorithm for dynamic control of robot manipulator is proposed. In this algorithm, information learned previously can be applied to new control objectives as long as similar regions of the system state space are involved. The control scheme requires no prior knowledge of the robot dynamics so that it is easy to be used to solve a particular control problem or to be modified to accommodate changes in the physical system. The control scheme is computationally efficient and well suited to fixed-point implementation. The effectiveness of the learning algorithm is verified by the computer simulation experiment of repetitive movements of two fore arms of 4 DOFs direct drive robot manipulator.

Key words: industrial robot cerebellar model learning algorithm CMAC mapping repetitive movement computer simulation

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自治式水下机器人导航与控制若干问题研究

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本文介绍了我国第一代自治式水下机器人导航及控制若干问题的研究工作。

具有某种程度自治能力的导航系统,必须拥有以下两个方面的能力:多传感器数据融合,在线故障监测。在本人的传感器融合算法中用了卡尔曼滤波技术,对潜水器的状态作出最优估计。我们用了冗余测量手段和假设检验方法实现故障监测。本文针对水下机器人控制问题,提出了两种鲁棒的自适应控制方法:一是多变量的自适应滑动控制方法,另一个是基于人工神经网络的新方法,称为有记忆的最小拍控制器,它类似于小脑模型连接控制器。

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